Causal Inference & Synthetic Control Engine for Algorithmic Pricing

Data Analytics Internship – Project Report

Dataset: Pricing_Dataset_4000_Rows.csv | Records: 4,000 | Products: Earphones, USB Cable

Prepared using the uploaded project dataset

1. Executive Summary

This report examines a synthetic pricing dataset containing 4,000 daily product records. The analysis compares Earphones (treatment group) and USB Cable (control group) around the stated intervention date of 15 May 2025. It includes descriptive analysis and a difference-in-differences (DiD) calculation, supported by six different chart types.

2. Dataset Overview

The dataset contains Date, Product, Category, Price, Units_Sold, Price_Change, Treatment_Group, Control_Group, Post_Period, and Revenue. The uploaded file contains 4,000 rows and 11 columns, covering 2024-01-01 to 2029-06-22.

3. Methodology

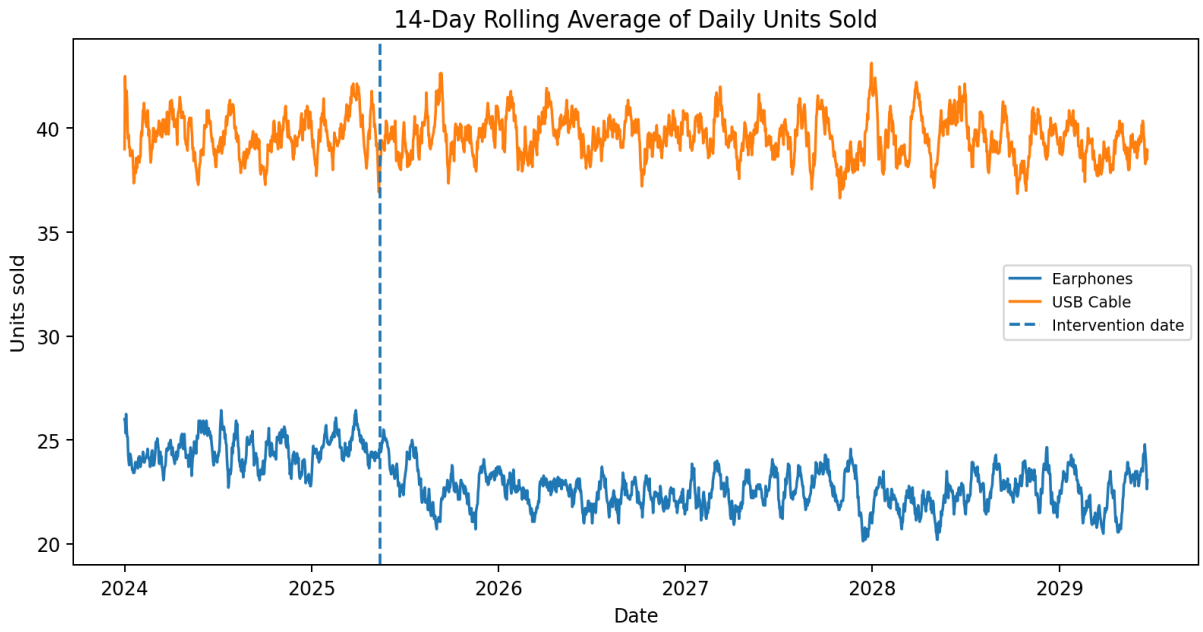
The workflow includes loading the CSV, checking the dataset structure, grouping records by product and period, calculating descriptive averages, visualizing trends and distributions, and calculating DiD as (treatment after – treatment before) – (control after – control before).

4. Descriptive Results

Average units sold per record: Earphones before = 24.49, after = 22.54; USB Cable before = 39.64, after = 39.59. The calculated DiD estimate is -1.901 units per record.

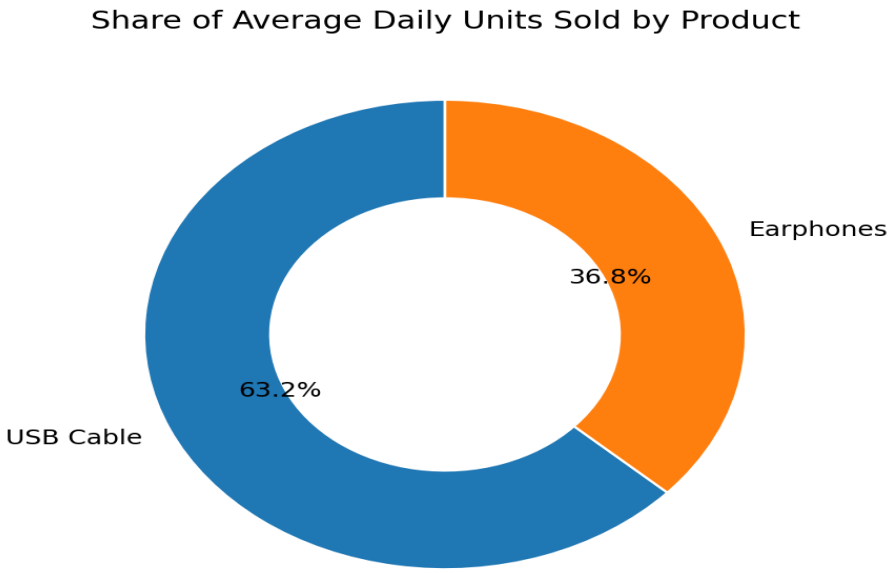
Product	Period	Mean units	Mean price (■)	Mean revenue (■)	Rows
Earphones	After	22.54	540.00	12171.60	1500
Earphones	Before	24.49	500.00	12245.00	500
USB Cable	After	39.59	200.00	7918.27	1500
USB Cable	Before	39.64	200.00	7928.00	500

5. Daily Sales Trend – Line Graph



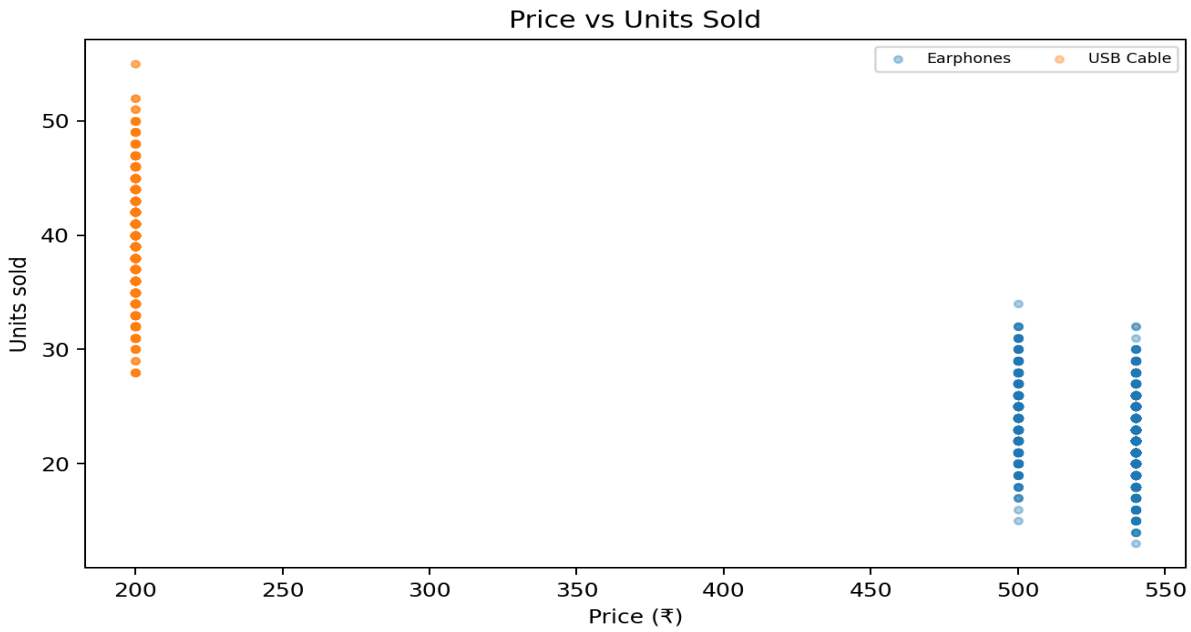
The line graph shows the 14-day rolling average of daily units sold for each product. The vertical dashed line marks the intervention date.

6. Product Contribution – Donut Chart



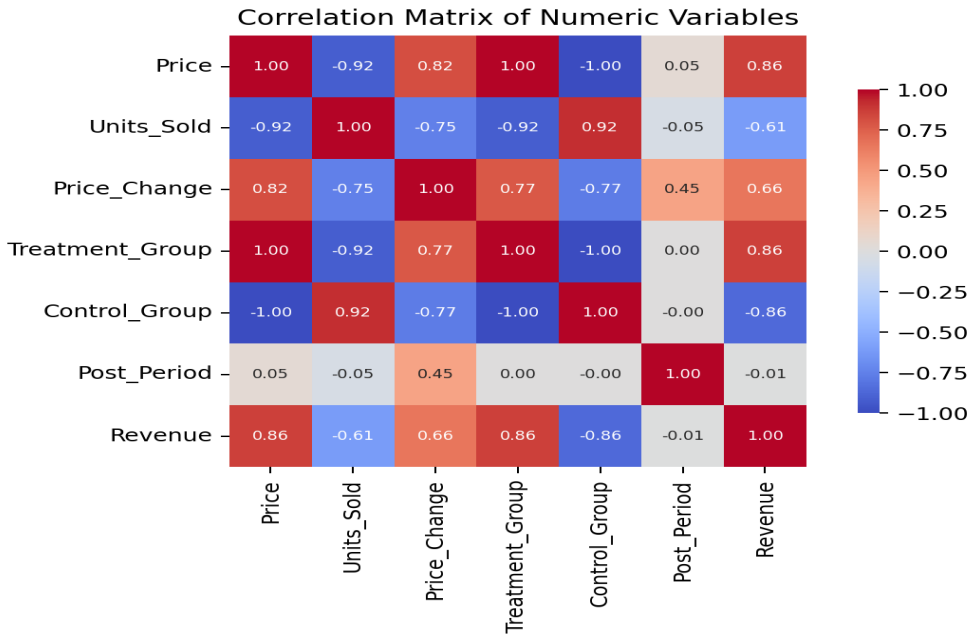
The donut chart compares each product's share of average units sold per record. It is a descriptive comparison, not a measure of causal impact.

7. Price and Demand – Scatter Plot



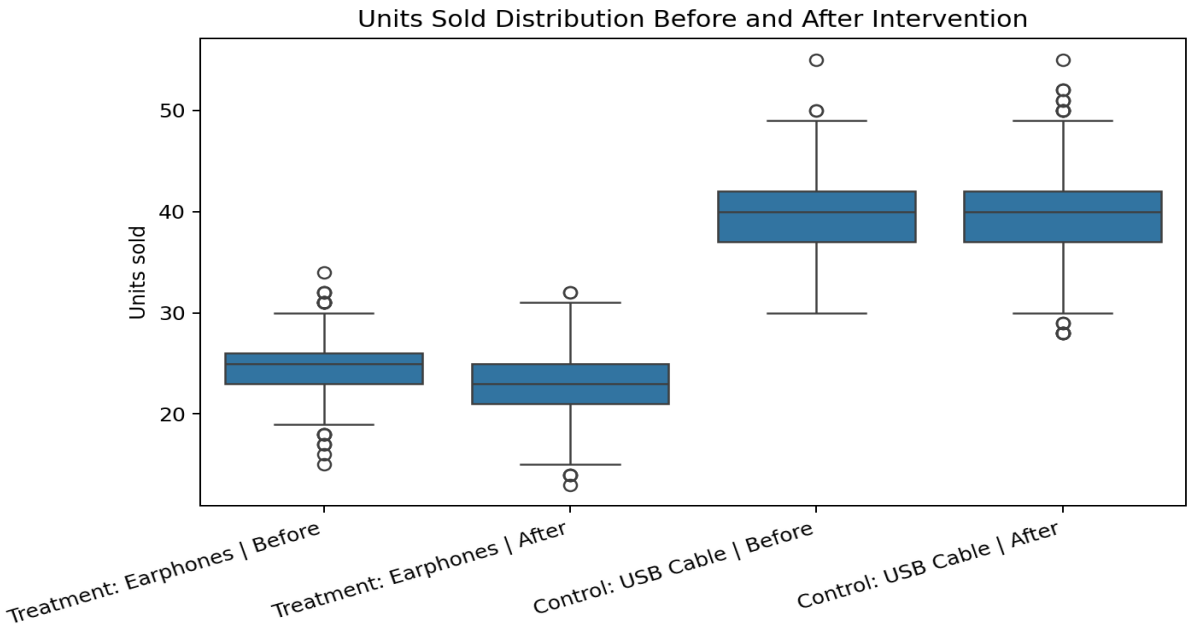
The scatter plot displays the relationship between listed price and units sold. The pattern should be interpreted descriptively because other factors may affect sales.

8. Numeric Relationships – Correlation Heatmap



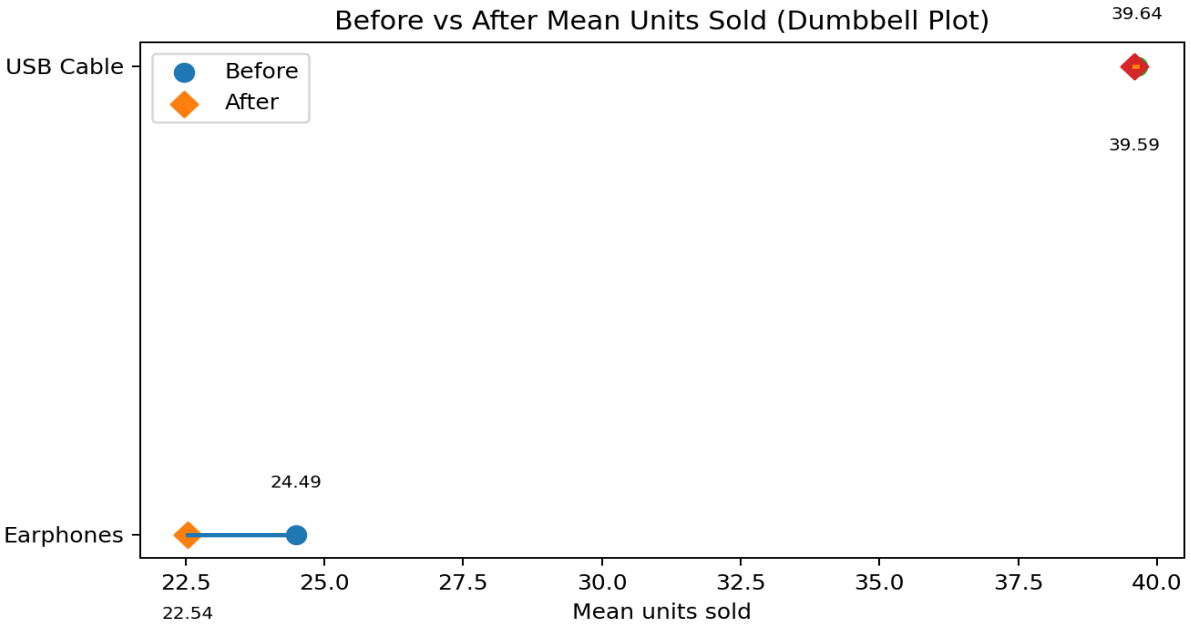
The heatmap summarizes pairwise correlations among numeric dataset fields. Correlation does not establish causation.

9. Sales Distribution – Box Plot



The box plot compares the distribution of units sold before and after the intervention for treatment and control products.

10. Before-and-After Comparison – Dumbbell Plot



The dumbbell plot connects each product's mean units sold before and after the intervention.

11. Interpretation and Limitations

The DiD estimate is a comparison of changes in the treatment and control series. It should not be presented as definitive causal proof. The dataset is synthetic, so its patterns are illustrative rather than evidence about an actual business. A causal interpretation would require credible treatment assignment, an appropriate comparison group, and a defensible parallel-trends assumption. Any synthetic-control donor series generated from the treatment series would also not be independent evidence.

12. Business Recommendations

Before changing prices in a real store, test pricing decisions on suitable real data, track units sold and revenue over time, compare against a defensible control group, and monitor customer and market factors that could influence demand. Use multiple metrics rather than units sold alone.

13. Conclusion

The project demonstrates a reproducible workflow for analyzing pricing data with descriptive statistics, visualizations, and a basic DiD comparison. The six chart types provide complementary views of trends, product contribution, price-demand patterns, correlations, distributions, and before-after differences. Findings remain illustrative because the dataset is synthetic.

14. Tools Used

Python, Pandas, Matplotlib, Seaborn, and ReportLab.